



ATRC Talent Platform

Functional Scope of Work

July 2026

Table of Contents

1.	ATRC Overview.....	4
2.	ATRC Talent Platform – Strategic Goals	4
3.	Project Overview	5
4.	Objective and Outcome.....	5
5.	In-Scope Deliverables	6
5.1.	Talent Website (Bilingual: EN/AR with full RTL)	6
5.2.	Mobile App (English, Native iOS + Android)	8
5.3.	Identity, Registration & User Profile (Website and Mobile App)	9
5.4.	Integration of ATRC Career Test (Website and Mobile App)	10
5.5.	Integrations (Website and Mobile App)	10
5.6.	Support & Warranty (Hypercare for Website and Mobile App).....	11
6.	Technical Requirements.....	11
7.	Use of AI, Automation, and Vibe Coding	11
7.1.	Overview.....	11
7.2.	AI-driven Development and Productivity.....	12
7.3.	AI-enabled Testing and Quality Assurance.....	12
7.4.	Vibe Coding Enablement (Citizen Development)	13
7.5.	Governance, Security, and Compliance of AI Usage	13
7.6.	Proof of Capability (Mandatory)	14
7.7.	Integration with ATRC Delivery Model.....	14
8.	Project Delivery Approach, Phases, and Location	15
9.	Acceptance Criteria (Go-Live Readiness)	17
9.1.	Overall Acceptance Criteria	17
9.2.	Website Architecture	18
9.3.	Registration & User Profile.....	19
9.4.	User Data Fields	19

9.5.	AI CV Builder	19
9.6.	Technologies and Pathways Sections.....	20
9.7.	Career Planner Section	20
9.8.	Talent Enablement Section	20
	10. Responsibilities	21
	10.1 ATRC Responsibilities	21
	10.2 Vendor Responsibilities	21
11.	Proposal Submission Requirements	23
12.	Evaluation Criteria	24

[Appendix 1. High-Level Sitemap](#)

[Appendix 2. Example of info provided by ATRC for and Advanced Technology Page](#)

1. ATRC Overview

The Advanced Technology Research Council (ATRC) is a UAE-based entity focused on building a world-class research and development ecosystem that strengthens Abu Dhabi's position as a global technology leader. Its talent enablement agenda is designed to identify, develop, and track R&D talent through integrated platforms, programs, and initiatives.

Through structured initiatives such as talent assessment, competitions, internships, and career pathways, ATRC accelerates the R&D talent pipeline and equips individuals with skills in advanced technologies.

Overall, the goal is to nurture local talent, support national innovation priorities, and contribute to the UAE's transition into a knowledge-based economy driven by advanced technology capabilities.

2. ATRC Talent Platform – Strategic Goals

2.1. Goal 1 – Provide useful tech- and career- resources for UAE Youth

ATRC aims to build a futuristic Talent Platform (website and a mobile app) for the UAE students (14 years old – 26 years old) that captures attention of the youth, presents advanced technologies in an appealing way and helps students to find their best career path and programs & tools to help them to develop their talent throughout their pathway from school to work.

View of success – to receive alike feedback from the target audience:

- “ As a UAE student from 14 y.o. to 22 y.o., I know this is a website to visit if you want info about STEM- and Future-related topics
- “ It looks so advanced – futuristic, flowing, fast, with all info nicely visualized and clearly explained. Totally 2025.
- “ It looks like a digital encyclopedia (in ENG and AR)
- “ I had no idea about what to for university – and this site was of huge help

Ideal: build it in a way to inspire students and parents share this info

2.2. Goal 2 – Contribute to Talent Management System

Talent Platform will serve as a talent data source for ATRC Talent CRM. It implies that Platform should have correct web analytics across webpages and app and extensive data

capabilities regarding user data collection (use interface is planned to have ~100 indicators).

3. Project Overview

ATRC seeks a qualified web development vendor with AI-tool building capabilities who works in Agile manner and actively follows latest developments in Generative AI and Agentic Coding in particular. ATRC expects the vendor to use AI in development, yet the final UX/UI shall not look AI-generated. Experience with EdTech is a plus.

The vendor will need to design, build, integrate, test, and launch an integrated digital talent platform consisting of:

- A Bilingual Talent Website (English + Arabic, full RTL) with headless CMS, incl.
 - A Career Test integrated into the website and mobile experience (the career test already developed by another vendor)
 - A Student/User Profile (approx. 100 fields) for 4 different types of users with export to ATRC systems
 - ATRC Talent Enablement Program Directory
 - Interactive Sections about Advanced Technologies, Career Pathways, Career Builder
 - Other Pages
- A Mobile App (mirroring the key parts of the website + a habit tracker feature, iOS + Android)

The Website and the app must be delivered as a single cohesive experience with shared identity, consistent branding, and integrated data flows.

4. Objective and Outcome

The platform shall:

- Provide a modern, futuristic, youth-friendly website and app experience showcasing advanced technology domains and pathways.
- Enable talent identification and tracking through structured profiles, assessments, analytics, and exports that support ATRC CRM workflows.
- Ensure UAE-first compliance: data sovereignty, security, accessibility, and identity-first architecture.
- Be built on a tech stack ready for the AI era, allowing flexible experience and fast integration of new webpages in the future.

- Not contain AI-slop (both in visuals and copywriting).

5. In-Scope Deliverables

5.1. Talent Website (Bilingual: EN/AR with full RTL)

The vendor shall deliver a production-grade Talent Website with a headless CMS and sitemap (see Appendix 1) including:

1. Homepage / Landing Page
2. Advanced Technologies section: nine (9) technology pages with curated content and interactive storytelling (with appropriate fallbacks)
3. Career Pathways section: ten (10) pathways pages (role overview, study routes, next steps)
4. Talent Enablement section: program listings with filters/search, program detail pages, eligibility/timeline/requirements, and actions (Save/Apply)
5. Supporting pages: About, Contact, FAQ, Terms, Privacy, plus remaining sitemap templates and supporting flows
6. Articles/Posts capability enabling ATRC to publish new content in a non-templated manner (CMS-enabled)
7. Site search: Site-wide search across technologies, pathways, programs, and articles (EN/AR)
8. SEO foundations: semantic markup, CMS-managed meta/OpenGraph tags, XML sitemaps, and hreflang for EN/AR to support organic discovery by the target audience
9. Admin portal: CMS back office with role-based access (via Microsoft Entra ID + MFA), content workflows, Talent Enablement program management, user management, and data export screens
10. User Profile with ~100 data fields (Incl. AI-tool CV Builder enabling users to generate a CV based on the data fields they filled and AI questions helping to refine a CV – to be developed by vendor)

ATRC has already developed UX/UI concept and homepage for the Website, but the other sections need to be created by the vendor with the quality and visual appeal similar to the screenshot below:



Find the pathway that can shape the UAE's future.

ATRC Talent Enablement program helps students explore future-focused technologies, understand where their interests fit, and move from curiosity to action through careers, competitions, and study pathways.



Announcement

The future starts here. Our STEM programme empowers students to explore cutting-edge technologies, discover their strengths, and turn ideas into action. Join a new generation.



Signs of the future

Did you know, that ...



90%

of the world's data was created in the last two years, accelerating AI and machine learning.



50,000

new jobs are projected through Abu Dhabi's smart and autonomous vehicle push.



2050

is the UAE Net Zero horizon shaping clean energy, water, and sustainability pathways.

Our Mission

We want to empower the next generation of innovators to shape future world.

You're the next generation.

[About Us](#)

The website should be built on the data provided by ATRC (about 500-600 ppt pages, ENG and AR) and the High-Level Sitemap (see Appendix 1)

Name	Date modified	Type	Size
FFCP.2. Resumes (ENG)	3/13/2026 12:10 PM	File folder	
FFCP.2. ATRC_Inspiration to the Youth_for Website (AR)	11/5/2025 11:13 AM	Microsoft PowerPo...	5,225 KB
FFCP.2. ATRC_Inspiration to the Youth_for Website (ENG)	3/24/2026 1:03 PM	Microsoft PowerPo...	25,051 KB
FFCP.2. ATRC_Interactive_Career_Guide (AR)	1/27/2026 4:37 PM	Microsoft PowerPo...	113,400 KB
FFCP.2. ATRC_Interactive_Career_Guide_vPDF (ENG)	2/11/2026 1:23 PM	Microsoft PowerPo...	247,653 KB
FFCP.3. ATRC_Advanced Technologies (AR)	1/13/2026 2:39 PM	Microsoft PowerPo...	28,995 KB
FFCP.3. ATRC_Advanced Technologies (ENG)	5/7/2026 1:19 PM	Microsoft PowerPo...	27,801 KB
TAT. 3. Competitions Section (ENG Only)	4/16/2026 12:34 PM	Microsoft PowerPo...	509 KB

5.2. Mobile App (English, Native iOS + Android)

The vendor shall deliver a mobile app with:

- Login: UAE Pass + OTP
- Onboarding aligned to user profile (school/curriculum/subjects, etc.)
- Card-based dashboard surfacing technologies, pathways, programs, saved items, reports
- In-app career test experience with results + PDF report access
- Program directory access (save/apply)
- Deep links to mobile web for long-form content
- App includes a gamified habit tracker for Habit/goal/routine tracking with analytics and notifications for users
- Feature for development in 2027 (to be priced separately): AI-Buddy. App includes an AI-chatbot (based on system prompt and minimal RAG) aimed at development of critical thinking skills in students – by asking thought-provoking questions that move students' thinking to higher levels of thinking.
- Release to Apple App Store and Google Play, with UAE region availability configuration
- Language: App UI is English at launch; the app architecture shall support the addition of Arabic (full RTL) in a future phase without re-engineering

5.3. Identity, Registration & User Profile (Website and Mobile App)

The vendor shall implement:

1. User registration (email + password) with verification and account management (reset/change)
2. UAE Pass OAuth integration for web SSO; and UAE Pass SDK integration for mobile
3. Minors: age verification at registration and a parental/guardian consent flow for users under 18, in line with UAE PDPL requirements for processing minors' data
4. Account deletion: self-service account deletion and data erasure flow, meeting Apple App Store and Google Play account-deletion requirements and UAE PDPL
5. Session security: JWT, refresh token rotation, rate limiting on auth endpoints
6. User profile model (~100 indicators) aligned to ATRC's master field specification (provided by ATRC), including school/curriculum/subjects, projects, competitions, internships, saved programs, assessment history, etc.
7. Role-aware profile behavior (student/parent/educator/other) and parent ability to associate multiple children (where feasible).
8. Export capability to ATRC systems: CSV/XLSX exports in two logical datasets (e.g., user data and skills), and integration-ready outputs for CRM ingestion
9. Clarification: in-scope CRM integration is batch export (CSV/XLSX, two datasets) plus integration-ready field mapping to the ATRC CRM data model; near-real-time API integration with ATRC CRM can be quoted separately as an optional deliverable.

ATRC will provide a list of ~100 indicators (and source code of AI-generated prototype) for the user profile with required options and interactivity:

ATRC User Profile – Master Field Specification April 2025								
* Mandatory / must complete * Optional / missing profile * Conditional / required when parent condition met * System - auto-populated / read-only								
Section	Field Name	Field Type	Options / Values	Status	Shown To	Condition / Trigger	DB Field Name	Dev Notes
YOUR INTERESTS								
Subtitle: Drives events & programs recommendations and research center matching								
1	ATRC Advanced Technologies You are Interested In	Multi-select	Artificial Intelligence (AI) / Artificial Intelligence (non-technical) / Quantum Computing / Renewable & Sustainable Energy / Population & Space / Directed Energy / Autonomous Robotics / Secured Systems / Advanced Materials / Cryptography	Optional	All	Always	tech_interest[]	Student selects all that apply.
2	ATRC Sectors of Interest	Multi-select	Aerospace and Space / Food and Agriculture / Healthcare / Safety and Security / Sustainability, Environment and Energy / Transport / Finance / Media / Entertainment / Tourism	Optional	All	Always	sector_interest[]	Student selects all that apply.
ACADEMIC JOURNEY – SCHOOL								
Subtitle for Students: School, university, subjects and grades								
Subtitle for Parents: Your Child's school, university, subjects and grades								
Subtitle for Educator: School/ University you work at, and subjects you teach								
Subtitle for Other: School, university, subjects and grades								
3	School I am currently...	Choice of 3 buttons	In School In College/University Graduated	Optional	All		current_ed_status	When In University OR Graduated, show University / College section. Sync with uni_enrolled.
4	School School Name	Autocomplete + free text	900+ UAE schools preloaded (KHDA / ADEK / SPEA). Free text fallback if not found. If a user picks "Other", the data is not added to CRM (it should be previewed by an analyst first).	Optional	All		school_name	If not in list: store as-is, school_verified=false. Source: Schools sheet.
5	School School Emirate	Dropdown	Abu Dhabi / Abu Dhabi - Al Ain / Abu Dhabi - Al Dhafra / Dubai / Sharjah / Ajman / Ras Al Khaimah / Fujairah / Umm Al Quwain / Outside the UAE	Pre-filled + Optional	All		school_emirate	Abu Dhabi split into three sub-regions: "Abu Dhabi - Mainland" shortened to "Abu Dhabi". Pre-fill from registration when possible.
6	School Curriculum	Dropdown	Display in this order (w/o numbers): 1. MoE (UAE) 2. British 3. International Baccalaureate 4. American 5. Indian / CBSE 6. SABIS 7. Other Grades 1-7 (single option) Grade 8 / Year 9 Grade 9 / Year 10 Grade 10 / Year 11 Grade 11 / Year 12 Grade 12 / Year 13	Pre-filled + Optional	All		school_curriculum	Drives Fields 12, 13, 14 and all grade suggestion logic. Pre-filled, but can differ from registration if student changed schools.
7	School Grade / year	Dropdown	MoE (UAE) only: General Advanced Technical & Vocational (TVET / Applied)	Pre-filled + Optional	All		school_grade	Grades 1-7 stored as single value grade_1_7. Grades 8-12 stored individually. Pre-fill from registration.
8	School Stream / track	Dropdown – curriculum dependent	All other curricula: General STEM (Science, Technology, Engineering, Mathematics) Humanities & Social Sciences Business, Economics & Management Arts & Creative Industries Science & Physical Education British: hidden – no overall grade exists Other Curricula: _____ out of _____	Optional	All		school_stream	MoE shows 3 options. All other curricula show 6 standardised options. Reset when curriculum changes.
9	School Overall grade/score (Note below: Apply your scale - e.g. 3.6 out of 4.0 or 87 out of 100)	Free text		Optional	All	Field 9 = No AND curriculum = British. Hidden for British.	school_overall_score + school_overall_max	Do not show for British curriculum. There should be 2 columns with all grades mixed. For example, GPAs in different formats are stored in one column (school_overall_score), max GPAs are stored in the second column (school_overall_max).

5.4. Integration of ATRC Career Test (Website and Mobile App)

The vendor shall integrate a career test featuring:

- Branching logic
- Results persistence to profile
- Results views (summary + detailed recommendations)
- Downloadable personalized PDF report
- Export capability (CSV) and CRM linking pipeline
- Results Saved in User Profile
- Integration boundary: the career test engine (question logic, scoring) is developed and hosted by the incumbent test vendor. The Vendor's scope is embedding/integration via the test vendor's documented API, ingestion of results into the user profile, PDF report rendering, and CRM linking. ATRC will facilitate access to the test vendor's API documentation and a test environment; the test vendor's own effort is outside this SoW.

5.5. Integrations (Website and Mobile App)

Vendor shall implement and test integrations including:

1. UAE Pass (web + mobile SDK)
2. Microsoft Entra ID + MFA for administrators (admin portal access control)
3. SMS/OTP provider (e.g., Twilio or ATRC-approved equivalent)

4. LMS integration from LearnWorlds including institutional accounts, bulk onboarding, campaigns, reporting dashboards
5. ATRC CRM export pipeline (CSV/XLSX outputs + integration-ready mapping)
6. Transactional email service (account verification, password reset, notifications) and mobile push notification infrastructure - providers subject to ATRC approval and UAE data-residency review.

5.6. Support & Warranty (Hypercare for Website and Mobile App)

Following Production Go-Live, the Vendor shall provide a warranty and hyper care support period as defined within the contract.

The warranty and hypercare period shall be a minimum of six (6) months from Production Go-Live of each phase, unless otherwise agreed in the contract.

During this period, the Vendor shall:

- Resolve application defects.
- Support production incidents related to the application.
- Address security vulnerabilities identified within the application layer.
- Provide deployment support for approved fixes.
- Support disaster recovery testing and validation activities.
- Participate in operational review meetings as required.

Any support requirements beyond the agreed warranty period shall be governed through a separate support and maintenance agreement.

6. Technical Requirements

Please refer to a separate document “2. ATRC Talent Platform - Technical SoW”
Indicative scale: For sizing and licensing purposes, vendors shall assume up to 100,000 registered users and 5,000 concurrent sessions in Year 1 (to be validated at design stage).
Final volumetrics will be confirmed during solution design.

7. Use of AI, Automation, and Vibe Coding

7.1. Overview

The Vendor shall demonstrate extensive and practical use of Artificial Intelligence (AI), automation frameworks, and modern development paradigms to accelerate delivery,

improve quality, and enable collaborative development models aligned with ATRC's innovation vision.

Vendors must position AI as a core enabler across the full software lifecycle, including design, development, testing, deployment, and optimization. Yet the final UX/UI should not look AI-generated (for both visuals and copywriting).

7.2. AI-driven Development and Productivity

The Service Provider shall:

- Demonstrate comprehensive use of AI tools to:
 - Accelerate frontend and backend development
 - Assist in code generation, refactoring, and optimization
 - Support UI/UX prototyping and design iterations
- Provide examples of:
 - AI-assisted development workflows
 - Reduction in development timelines achieved through AI
- Clear outline:
 - AI tools/platforms used (e.g., IDE copilots, code assistants, design AI tools)
 - Governance mechanisms for AI-generated code quality and maintainability

7.3. AI-enabled Testing and Quality Assurance

The Service Provider shall leverage AI to enhance quality assurance by:

- Implementing AI-assisted:
 - Test case generation
 - Automated regression testing
 - Defect prediction and root cause analysis
- Enabling:
 - Continuous testing pipelines integrated with CI/CD
 - Intelligent monitoring and anomaly detection
- Demonstrating measurable improvements in:

- Test coverage
- Defect leakage reduction
- Release cycle time

7.4. Vibe Coding Enablement (Citizen Development)

The Service Provider shall enable a “Citizen Development” model within ATRC, allowing non-traditional developers (citizen developers) to actively contribute to application development through no-code/low-code tools.

This shall include:

- Provision of:
 - AI-assisted low-code / no-code / vibe-coding environments
 - Prompt-driven or natural-language-driven development interfaces
- Support for:
 - Rapid prototyping by ATRC business users and SMEs
 - Creation of simple modules, workflows, and UI components by non-engineers
- Implementation of:
 - Guardrails for secure and compliant citizen development
 - Validation pipelines before promoting vibe-coded components to production

Appropriate validation and governance mechanisms shall be implemented to ensure quality and security before deployment.

7.5. Governance, Security, and Compliance of AI Usage

The Service Provider shall ensure that all AI usage complies with ATRC and UAE regulatory requirements:

- Adherence to:
 - Data residency requirements (UAE-hosted AI processing where applicable)
 - ATRC AI tool usage policies and cybersecurity guidelines

- Implementation of:
 - Secure handling of prompts, source code, and user data
 - Strict controls to prevent leakage of confidential information via AI tools
- Provision of:
 - Audit trails for AI-generated outputs
 - Explainability mechanisms (where applicable)

7.6. Proof of Capability (Mandatory)

The Service Provider must provide demonstrable evidence of AI and vibe coding capabilities, including:

- Case studies showing:
 - End-to-end use of AI in software delivery
 - Quantifiable productivity improvements
- Live demonstrations and/or prototypes:
 - AI-assisted development workflows
 - Vibe-coded modules or applications and citizen development enablement.
- Access (if possible) to:
 - Trial environments or sandbox demos

Submissions lacking clear and validated evidence of AI-driven delivery and vibe coding capabilities may be deemed non-compliant.

7.7. Integration with ATRC Delivery Model

The proposed solution must:

- Aligning with ATRC's broader AI strategy and platform ecosystem
- Be compatible with ATRC's cloud, security, and DevOps standards
- Enable future scalability for AI-driven features within the STEM platform (e.g., AI CV tools, recommendation engines, conversational interfaces)

8. Project Delivery Approach, Phases, and Location

Vendor shall execute a two-phase delivery with parallel workstreams, targeting a unified web + mobile launch (Agile-Waterfall approach).

Entire Project: 3 phases

- Phase 1 focuses on website foundations
- Phase 2 adds deeper modules and converges with mobile app go-live. *ATRC acknowledges that publishing of mobile application requires approvals from app stores, and thus the deadlines might be extended.
- Phase 3 adds AI Buddy feature and “Citizen Development” modules
- Indicative timeline: Phase 1 within 3 months of contract signature; Phase 2 within 6 months, converging with mobile app go-live; Phase 3 during 2027 (priced separately). Vendors may propose an alternative schedule with justification.

For the future years of collaboration, we seek an Agile approach and a possibility of “Citizen development” allowing integrating new website sections produced with no-code tools into the main website. The code handed over by ATRC to be considered as a draft verified in terms of functionality, yet the Vendor would need to align the final design with the website overall and ensure that it meets ATRC IT requirements.

Please provide time and commercial estimations for an integration of one new section in the proposal and accommodation of “Citizen development” approach.

Please use this table to add mandays per milestone, prices, other information if needed:

#	Phase	Deliverable	Mandays, etc.
Talent Website			
1	Phase 1	Home landing page <i>Design developed and to be provided by ATRC in Figma</i>	
2	Phase 1	9 Advanced technologies	

#	Phase	Deliverable	Mandays, etc.
		Source: FFCP.3. ATRC Advanced Technologies (to be provided)	
3	Phase 1	Integration of Career test <i>Interactive quiz is developed by an external vendor</i>	
4	Phase 1	Registration & About ATRC	
5	Phase 2	User account <i>~100 indicators about user academic and work journey, skills. functional prototype developed, some edits might be required, UX/UI required</i>	
6	Phase 2	AI-tool CV Builder <i>Helps to build a CV based on the filled fields in the use account and questions helping to improve the CV, with a Word and PDF download option</i>	
7	Phase 2	Career Pathways <i>Source: FFCP.2. ATRC_Interactive_Career_Guide (ENG & AR) (to be provided)</i>	
8	Phase 2	Talent Enablement Programs <i>ATRC Talent Enablement Initiatives and Info about them</i>	
9	Phase 2	Career Planner <i>Hidden for unregistered users · Source: FFCP.2. ATRC_Interactive_Career_Guide (to be provided)</i>	
10	Phase 2	Integrations (LMS module, UAE Pass)	
Mobile Application			
11	Phase 2	Mobile Application Development and (potentially) Publishing, incl. Habit Tracker feature <i>Habit tracker to help track study activities and develop habits</i>	

#	Phase	Deliverable	Mandays, etc.
Additional Scope			
12	Phase 3	Development of AI-tool AI Buddy <i>AI tool based on a system prompt + some RAG to improve critical thinking of students</i>	
13	Phase 3	Integration of new sections ("Citizen Development")	

Final milestone plan shall be proposed by vendor, but must accommodate third-party approval lead times (UAE Pass approvals, Apple/Google reviews).

Location: Team members can be based outside of the UAE, but all IT specialists who will have access to ATRC systems need to pass security clearance and have several IT members present in the UAE during go-lives (to be included in the cost).

9. Acceptance Criteria (Go-Live Readiness)

9.1. Overall Acceptance Criteria

- The website meets all agreed project specifications and functionalities.
- The website is responsive, bilingual (English/Arabic) with an integrated career test, generates reports and has analytics integrated across all pages.
- All links, buttons, forms, and integrations work correctly without errors.
- Design matches approved mockups (layout, colors, fonts, images).
- Content is complete, accurate, free of typos, and avoids typical AI-structured phrases like:
 - "It's not X, it's Y"
 - Intro → 3 bullet points → conclusion pattern
 - Em dashes ("-")
 - "Signal", "noise"
- The design of the website is unique and fits ATRC Brand style and ATRC quality of visual materials.
- The website and app design does not look like a ready template / AI-generated website in terms of visuals and copywriting.

- All AI-generated pictures are of high quality and do not look like AI slop (blurred faces, weird details, etc.).
- UX/UI guides attention in a nice way. Visual priority is given to the most important information, explanatory is presented in a smaller font.
- The website loads quickly and is fully responsive on desktop, tablet, and mobile. Website pages shall be optimized to load within an average target of 2.5 seconds under normal UAE internet conditions (smartphone devices, iOS / Android, across top-5 browsers), subject to final content and hosting environment. Dependent factors like local internet ping times and DNS resolutions are outside of Vendor's scope.
- Works properly on major browsers (Chrome, Safari) and agreed devices.
- Uses HTTPS and includes proper security measures and input validation.
- All critical and major bugs found during testing are fixed before acceptance.
- UAT process: ATRC performs UAT with Vendor support; defects are classified Critical / Major / Minor with agreed fix SLAs; up to 3 design revision rounds per deliverable are included in the fixed price.
- The Website seamlessly integrates the Career Quiz developed by Another Vendor, and results of the career quiz are saved in the user profile and can be downloaded as .csv and linked to ATRC CRM.
- Quality assurance is diligently performed at Vendor side. Vendor ensures that content provided is implemented accurately without typographical or formatting errors. The responsibility for translation accuracy remains with the ATRC provider (translation is provided for the content only, not for the website titles and user fields).
- Back-end is user-friendly; the CMS enables ATRC to perform standard content edits independently. Structural or advanced changes may require Vendor support (target: not more than 5% of content). Only advanced structural / module-level changes require Vendor involvement.
- Data between frontend and backend shall remain consistent to the extent defined in the approved information architecture and scope of work.

9.2. Website Architecture

- The website is built according to the Approved High-Level Sitemap (Appendix 1) plus a section with Articles (so that ATRC can add an article) in a non-templated way.
- The website is built in a way so that new pages can be added quickly.
- An ATRC-approved analytics solution is added to each page so that ATRC can track visits and key actions; any analytics processing of user data outside ATRC-approved UAE regions (e.g., Google Analytics) requires prior ATRC Security & Compliance

approval, consistent with the Technical SoW. A tracking plan (event taxonomy for key actions) shall be agreed before build.

- Save / Apply buttons are included.

9.3. Registration & User Profile

- Built in this format: ATRC Student Profile.
- Uses the info from file: User_Data_Fields_ATRC_final (~100 indicators).
- Visually looks more like current UX/UI (purple background and gradient buttons).
- Can be filled quickly: if there are breakdowns of 2-4 options (example: male / female and role: student / parent / educator / other), they are shown as buttons the user can tap quickly.
- Pre-filled as much as possible.
- User profile is adjusted to the roles: student / parent / educator / other (not that many adjustments at this point though, for example, educators fill in the data for the school / university / college at which they work, not study).
- A parent can add several children (if possible).
- Data (~100 indicators) can be exported to CRM in 2 tables: (1) User data and (2) Skills, in .xlsx or .csv.

9.4. User Data Fields

The full ATRC User Profile Master Field Specification (~100 indicators) to be provided along with the source code (developed by career counselors with AI).

ATRC User Profile — Master Field Specification April 2026									
* Mandatory = must complete * Optional = enters profile * Conditional = required when parent condition met * System = auto-populated / read-only									
Section	Field Name	Field Type	Options / Values	Status	Shown To	Condition / Trigger	DB Field Name	Dev Notes	
YOUR INTERESTS									
Subtitle: Dives events & programs recommendations and research center matching									
1	ATRC Advanced Technologies You are Interested In	Multi-select	Artificial Intelligence (ML) / Artificial Intelligence (non-technical) / Quantum Computing / Renewable & Sustainable Energy / Population & Space / Directed Energy / Autonomous Robotics / Secured Systems / Advanced Materials / Cryptography	Optional	All	Always	tech_interest[]	Student selects all that apply.	
2	ATRC Sectors of Interest	Multi-select	Aerospace and Space / Food and Agriculture / Healthcare / Safety and Security / Sustainability, Environment and Energy / Transport / Finance / Media / Entertainment / Tourism	Optional	All	Always	sector_interest[]	Student selects all that apply.	
ACADEMIC JOURNEY — SCHOOL									
Subtitle for Students: School, university, subjects and grades									
Subtitle for Parents: Your Child's school, university, subjects and grades									
Subtitle for Educator: School/ University you work at, and subjects you teach									
Subtitle for Other: School, university, subjects and grades									
3	School I am currently...	Choice of 3 buttons	In School In College/University Graduated	Optional	All		current_ed_status	When In University OR Graduated, show University / College section. Sync with uni_enrolled.	

9.5. AI CV Builder

- Fully integrated on the website (not as an external tool), meaning that it uses Registration and User Profile fields as inputs instead of offering another set of fields to fill in.

- Offers 5-7 modern-looking designs with several color and font options.
- Fits the CV design for different professionals (students, working professionals, PhDs, etc.).
- CV builder is intelligent and suggests improvements (there is value added on top of what free tools are offering, for example, ChatGPT, Enhancv, or CVpop).
- Adapted to the UAE best practices.
- Allows export in .pdf and .docx.
- System prompts are shared with ATRC.
- Automatic fill-in from a ready CV in PDF format - when a user uploads a CV / LinkedIn Profile Summary (downloadable as PDF) in PDF, the Profile fields will be pre-filled for the user's approval.

9.6. Technologies and Pathways Sections

- Look futuristic, with interactive elements and visuals.
- Not text-heavy: information appears when a user clicks on the element.
- Based on content provided by ATRC.

9.7. Career Planner Section

- Meets the requirements for Technologies and Pathways Sections.
- Career Planner Section is visible to registered users only; other users can see that it is hidden and they need to register to unlock it.

9.8. Talent Enablement Section

- Section shows ATRC initiatives across the Researcher Journey (design developed by the previous web developer).
- For the School Program integration, Vendor will purchase and integrate the LMS at www.learnworlds.com. Please budget at USD 450 per month (from integration date for one year).
- ATRC can add the initiatives easily, ideally in a visually appealing form so that initiatives look like landing pages.
- This is the information requested for the high-level descriptions of Talent Enablement initiatives:
 - Name of the Initiative
 - Target age / grades
 - Short description
- After a user clicks on the initiative, the following will be shown:
 - Picture

- Name of the Initiative
- Target age / grades
- Short description
- Detailed description of the initiative
- Dates (if applicable)
- Place (if applicable)
- Deadline (if applicable)
- Location
- Apply button (saved in the user profile)
- Save button (saved in the user profile)

10. Responsibilities

10.1 ATRC Responsibilities

- Requirements validation & sign-off
- Provide content, translations, and brand assets
- Deliver final source content (approx. 500–600 PPT pages, EN and AR) per an agreed content-delivery schedule aligned to the project plan; content readiness is a critical-path dependency
- Ensure SME support from the vendor who is well informed about ATRC content and the project overall
- Own Azure subscriptions, billing, domain/DNS, and certificates
- Provide credentials/keys for the UAE Pass approvals (web + mobile)
- Provide/approve Apple and Google developer accounts under ATRC organization for publishing

10.2 Vendor Responsibilities

- Solution architecture & design
- UX/UI design (web & mobile)
- Talent website development
- Content transformation, entry, and structuring in the CMS based on ATRC-provided source materials (including EN/AR versions)
- Mobile app development (iOS/Android)
- Career test integration
- User profile (100 fields) UX/UI design & build

- Data model & API integration with ATRC systems
- Provide CI/CD, application configuration, and deployment support
- Implement monitoring/telemetry and security controls aligned to scope requirements
- Advanced tech content experience
- Testing (SIT/UAT support)
- Provide documentation, handover, and knowledge transfer
- Ongoing support & maintenance

11.Proposal Submission Requirements

Potential vendors are invited to provide a concise proposal consisting of Product, IT, and Commercial Parts including:

1. Product Part (up to 20 pages):
 - 1.1. Overview and Key Value Proposition of the company, key differentiators from competitors (short description)
 - 1.2. Project Timeline (in 3 phases)
 - 1.3. Deliverables and Price: a table with deliverables (incl. short description), prices and man-days (broken down by deliverables)
 - 1.4. Clear and Transparent Scope Inclusions and Exclusions (also included in Commercial Part)
 - 1.5. Project Team: Position, experience, involvement in the project (X / 8 hours a day on average)
 - 1.6. Portfolio: A Table with location (UAE/Outside), client type / name (if can be revealed) + Link to website / App name
 - 1.7. Visual Examples: 1 page with screenshots of the best and most relevant works
 - 1.8. Considered as a plus: high-level prototype / design of the potential website or app
2. IT Part (any volume of submitted documents):
 - 1.9. Overview and Key Value Proposition of the company, key differentiators from competitors (short description)
 - 1.10. Technical Infrastructure & Cloud Hosting
 - 1.11. How AI is used throughout the Project
3. Commercial Part (broken down by deliverables and showcasing scope inclusions and exclusions. Please use the table above in Section 8)

12.Evaluation Criteria

Evaluation Criteria	Description	Weight
AI & Innovation Maturity	Team's expertise in development of innovative digital products, ability to successfully execute projects of similar scope and complexity (websites, apps, AI chatbots, AI apps)	20%
Technical Capability	Evaluation of the proposed technology architecture and ability to deliver “citizen development” approach	25%
Design & UX Quality	Overall quality of Digital Products in the Vendor Portfolio and/or submitted Talent Platform’s design / prototype	10%
Functional Scope Coverage	Demonstrated understanding of the project. Clarity and transparency of the scope	25%
Delivery Approach and Timeline	Proposed project plan, milestone clarity, schedule feasibility, resource allocation, and the team's ability to deliver within the defined timeframes	20%
Total		100%

APPENDIX 1.

ATRC Talent Platform High-Level Sitemap (The sections below are for the website. Mobile app sections to be simplified by the vendor). The sections are marked as Phase 1 or 2 depending on the release.

1. Home Landing Page (Phase 1)

- Primary Action: Take a quiz
- Secondary Action: Login
- Hero Section: Design Your Future
- Intro Section: Future Statistics
- Our Mission Section: Main Statement (CTA About us)
- Section: Talent Enablement
 - Action: Explore ATRC Programs →
 - Subpage: ATRC Talent Enablement (Program Listing)
- Career Paths: Career Paths Overview (Keep this in a Grid view)
 - Action: More About Career Paths →
- Advanced Technologies Section: Technologies Overview
 - Action: Learn more about technologies → 2. Advanced Technologies
 - (On click on technology) Action: Explore Technology → Subpage: Specific Technology Details
- Featured Spotlights: Future Heroes Slider (Information source: FFCP.2. ATRC_Inspiration to the Youth_for Website (ENG and AR)
- Call to Action Section: Prompt to take a quiz
 - Primary Action: Take a quiz
- Footer

2. Advanced Technologies (Information Source: FFCP.3. ATRC_Advanced Technologies) (Phase 1)

- Overview Section (Introduction)
 - Why Important?
- Listing Section: All Advanced Technologies
 1. AI and Digital Science (STEM)
 2. AI and Digital Science (Non-STEM)
 3. Quantum
 4. Renewable & Sustainable Energy
 5. Propulsion & Space
 6. Directed Energy

- 7. Autonomous Robotics
- 8. Secure Systems
- 9. Advanced Materials
- 10. Cryptography
 - Biotechnology (Not Visible on the listing page)
- (On click on technology) Action: Explore Technology → Subpage: Specific Technology Details

3. Career Pathways (Information source: FFCP.2. ATRC_Interactive_Career_Guide (ENG and AR) (Phase 2))

- Action: Take Career Quiz →
- Intro Section:
 - Why to think about future path?
- Explanatory Section:
 - Steps: What are career paths
- Listing Section: Pathways
 - Future Builder: Engineers
 - Future Explorer: Scientists
 - Future Analyst: Data, Finance
 - Future Creator: Design, AR/VR
 - Future Leader: Entrepreneurs, Managers
 - Future Strategist: Advisors, Policy Developers
 - Future Communicator: Media, Education
 - Future Healer: Medicine
 - Future Sustainability Champion: Sustainability
 - Future Enabler: Law, HR, Security
- (On click on career pathway) Action: Explore Career Pathway → Subpage: Pathway Detail Content
 - Overview Section: Who they are, what they do
 - Career Section: Cool careers to explore
 - Skill Stack: Technical & Soft Skills
 - Roadmap Section: Academic & Career Roadmap
 - Degree Roadmap: Bachelors → Masters → PhD
 - Subject Mapper: High School Subjects
 - Section: Where to Study: UAE & Global Universities
 - Section: Future Skills to Develop

- Section: Profile Building
- Section: Professions
 - (On click on profession) Action: Learn more → Job Descriptions
- Key Subjects to study
- Technology Video(s)
- Competition Links
- Suggested Technologies (Other Technologies)
- Action: Take Career Quiz →

4. Talent Enablement (Phase 2)

- A visual with 6 Stages of a Researcher Journey
 - (On click on Each stage) Action: Explore Stage →
 - Stage Detail page with ~5 ATRC initiatives under each Stage

5. Career Test (Interactive Quiz developed by another company) (Phase 1)

- Action: Take Career Quiz →
- Dashboard

Note: The results will be stored under the user account dashboard, we'll show the primary persona match as well as the secondary matches. A downloadable pathway report will also be available in the user account. From that report, we can connect users to "3. Career Pathways"

6. Career Planner (hidden for unregistered users) (information source: FFCP.2. ATRC_Interactive_Career_Guide) (Phase 2)

- Sign-up is required (Prompt to sign-up)
- 7 Sections with PDFs that can be scrolled on the same page with info:
 - *Overview: Step by step Guide (Plan with tasks)*
 - *Action: Take Career Quiz →*
 - *Profile Building Overview*
 - *What is Profile Building?*
 - *Why the subjects matters?*
 - *Subjects & Exams*
 - *Mistakes to Avoid*
 - *From Subjects to Pathways*
 - *Grade Planner: Grades 9-12 Timeline*
 - *Focus & Grow*
 - *Outcome*

- *AI prompts (information source: 4. Career Report)*
- *Resources*
 - *Courses & Programs*
 - *Projects & Challenges*
 - *Research Platforms*
 - *Useful Materials*
- *Showcasing Profile*
- *CV & Portfolio Builder*
 - *CV Writing Tips*
 - *LinkedIn Basics & Tips*
 - *Impact-Based CV Examples*
 - *Portfolio Tools: GitHub, Behance, Notion*
- *University Prep*
 - *UAE-Specific Admissions & Requirements*
 - *Global University Pathway Requirements*
 - *Personal Statement Tips*
 - *Exam Milestones: EmSAT, AP, A-Levels*
 - *Action: Build Your CV →*
 - *CV & Portfolio Builder*

7. LMS Module (Phase 2)

- Integrated LMS module from LearnWorlds.com (with SSO with the main website)

8. About (Phase 1)

- About Us: Mission & Vision 2071
- ATRC (External Link)
- Useful Materials
- For Educators
- Contact: Email & Support Form
- Privacy & Terms (in the footer)

9. User Account (Phase 1: registration & core profile; Phase 2: full ~100-indicator profile & AI CV Builder)

- Logged In / Login / Register
- User Profile (~100 fields – data and source code to be provided by ATRC)
- Saved Career Pathway Results (integration with another vendor's solution)
 - Primary Career Pathway

- Secondary Career Pathways
- Saved Items: Courses, Clubs, Competitions

APPENDIX 2.

Example of info provided by ATRC for an Advanced Technology Page (Quantum)

Quantum

[Link](#)
[Back](#)



Quantum physics, also known as quantum mechanics, is a branch of physics that deals with the behavior of particles at the smallest scales. Quantum is all around us and we all are quantum.

WHY QUANTUM IS CRUCIAL FOR THE FUTURE

- **Quantum discoveries** have significantly shaped our economy, as, for example, lasers for communication and health, magnetic resonance for MRI, atomic clocks for GPS and semiconductors for the computing industry.
- **Quantum principles** underpin the way information is encoded, processed and transmitted, both in classical and quantum communication.
- **Quantum computers**, though still in early stages, hold immense promise. Unlike classical computers that use bits (0s and 1s), quantum computers use qubits, which can exist in superpositions. This allows them to solve certain problems exponentially faster. For example, they could revolutionize drug discovery, optimize supply chains, and tackle complex simulations.
- **Quantum-resistant cryptography** is crucial. While quantum computers are years away from breaking current encryption, it's essential to develop and implement quantum-resistant encryption now. Protecting our communications infrastructure is a priority.

KEY SUBJECTS YOU WILL STUDY

- **Quantum Mechanics:** Understand wave functions, probabilities, and the Schrödinger equation.
- **Quantum Computing:** Learn about qubits, gates, and quantum algorithms (like Shor's algorithm for factoring).
- **Quantum Information Theory:** Explore quantum entanglement, teleportation, and quantum communication.

12

Professions in Quantum



Quantum

Type	Complex of Professions
Specialized Professional	Quantum Researcher, Quantum Software Developer, Quantum Hardware Engineer, Quantum Key Distribution Engineer, Post-Quantum Cryptography Engineer
Requires Master / PhD	
Universal Professional	Optical Engineer, Electronics Engineer, FPGA Engineer, DevOps Engineer, Web Developer, UI/UX Designer, Quantum Analyst, Technical Project Manager, Solution Architect, Software Engineer
Requires Bachelor / Master	
Supporting role	Quantum Hardware Technician
Requires up to College / Bachelor	

Who/What professions are there in the UAE?

- Engineer (Mathematics, Data analysis, Software Programming)
- Quantum and Topological Systems
- Graduate Tutoring - Computer, Software Engineering, Mathematics, Physics
- Research Scientist in the Division of Engineering - Electrical and Computer Engineering
- Open Rank Faculty - Physics
- Research Scientist in the Division of Science - Mathematics
- Post-Doctoral Associate in the Center for Quantum and Topological Systems
- Research Scientist in the Division of Science - Mathematics, Physics
- Principal Engineer (Electronics Engineering)
- Senior Principal FPGA Design Engineer - Quantum
- Research Assistant in the Division of Engineering - Computer Engineering
- Research Assistant in the Division of Science - Physics
- Research Associate in the Center for

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36

Technology Priority: Quantum



Domain	Technology Goal	Education
Algorithms	Creating smart instructions (called algorithms) that help quantum computers solve very hard problems - like finding patterns in data or simulating how molecules behave. These are the brains behind future supercomputers.	
Physics	Studying how tiny particles (like atoms) behave when they're condensed in strange ways (superconductivity). This helps us build cool things like quantum gas systems and super-sensitive clocks.	Master in Physics, Mathematics or Engineering
Middleware	Building the "middle" that developers use to write quantum programs, simulate quantum computers on regular machines, and control quantum devices. Think of it as the bridge between ideas and machines.	Bachelor in Physics, Mathematics, Engineering
Communication	Designing futuristic communication systems that use quantum particles to send messages securely. This includes developing lab experiments like tiny chips and creating special light particles for quantum internet.	PhD in Quantum Science (Theory or Experimental)
Computing	Building the first quantum computer in the MENA region! It works at super cold temperatures, so quantum effects don't disappear. This is where quantum magic becomes real.	
Sensing	Making ultra-sensitive sensors that can detect tiny changes in magnetic fields or movement. These sensors use floating particles in vacuum and are key for scientific discoveries and medical technology.	

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37

Quantum – Currently Offered and Recommended Programs



Current offerings

Program	Bachelor	Master	PhD
Current Programs	Physics "Any Engineering Program with Mathematics" Applied Mathematics Electrical Engineering Computer Engineering Computer Science	Physics Mathematics Applied Mathematics Computer Engineering Computer Science Data Science Electrical and Computer Engineering	Physics Computer Science Data Science Electrical and Computer Engineering
Recommendation to seek training in	Quantum Algorithms Quantum Physics Quantum Middleware Quantum Communication Quantum Computing Quantum Sensing Electronics Engineering	Quantum Algorithms Quantum Physics Quantum Middleware Quantum Communication Quantum Computing Quantum Sensing	Quantum Algorithms Quantum Physics Quantum Middleware Quantum Communication Quantum Computing Quantum Sensing

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38

Quantum – Where you can study



Current offerings in Abu Dhabi

Program	Degree	University
Computer Engineering	Bachelor	Abu Dhabi University
		Xuabhi University
		NYU Abu Dhabi
	Master	USF University
		NYU London
		Xuabhi University
Computer Science	Bachelor	NYU Abu Dhabi
		USF University
		Xuabhi University
	Master	MSLUS
		NYU Graduate School of Arts & Sciences
		NYU London
PhD	MSLUS	NYU Abu Dhabi
		NYU Graduate School of Arts & Sciences
	PhD	NYU London
		NYU London

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39

Quantum – Where you can study

Current offerings in Abu Dhabi



Program	Degree	University
Ph.D.	Bachelor	Khalifa University
	Bachelor	NYU Abu Dhabi
	Bachelor	UAE University
	Bachelor	NYU Executive School of Arts & Business
	Bachelor	Khalifa University
Master's	Master's	NYU Abu Dhabi
	Master's	UAE University
	Master's	Khalifa University
	Master's	NYU Executive School of Arts & Business
	Master's	Khalifa University

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23

Postdoctoral / Senior Researcher in Quantum Device Characterization

III



Job Summary

Quantum system characterization naturally draws upon a diverse range of fields and methods, from information and learning theory, computer science, mathematical and theoretical physics, to signal processing, applied statistics, software development, hardware-efficient implementation, and experimental phenomenology. We are looking for candidates with expertise in any one of those fields and willingness to collaborate on both theoretical foundations and practical implementations of quantum characterization protocols and related topics.

Key Responsibilities

- Research and development towards scientific publications.
- Development and implementation of quantum characterization protocols in different experimental and theory collaborations.

Qualifications

- PhD degree in Physics, Mathematics, Computer Science, or a related field.
- International track record in research on Quantum Technologies, Quantum Information Theory, or related fields.
- Ability to work collaboratively in a multinational team environment, and also independently.

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24

Software Engineer – Quantum Computing Control

III



Job Summary

Quantum control requires expertise in quantum computing and control systems, and expertise in quantum computing and control systems. These roles are critical to the development of quantum computing systems. The roles require expertise in quantum computing and control systems, and expertise in quantum computing and control systems.

Key Responsibilities

- Design, develop, and maintain quantum control systems for quantum computing applications.
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- Develop and maintain quantum control systems for quantum computing applications.

Key Qualifications

- PhD degree in Physics, Mathematics, Computer Science, or a related field.
- International track record in research on Quantum Technologies, Quantum Information Theory, or related fields.
- Ability to work collaboratively in a multinational team environment, and also independently.

Key Skills

- Quantum computing and control systems.
- Quantum computing and control systems.
- Quantum computing and control systems.
- Quantum computing and control systems.

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25

Post-Doctoral Associate in the Center for Quantum and Topological Systems

New York University Abu Dhabi



Description

The Center for Quantum and Topological Systems (CQTS) at New York University (NYU) Abu Dhabi, UAE invites applications for a Research Associate / Post Doctoral Associate position in Quantum Computing. The candidate is expected to do cutting-edge research at the intersection of the following areas, therefore prior expertise in these topics is highly encouraged: Quantum Machine Learning, Machine Learning on Quantum Computers, Topological Data Analysis.

The selected candidate will work on cutting edge technologies in an excellent research environment, with a potential to work with a Quantum Computer through our collaboration partners. The Center possesses the unique possibility to investigate cutting-edge interdisciplinary questions within the theoretical and fundamental aspects of Mathematics, Physics, and Computer Science and Engineering, from various points of view and different abstraction levels. The selected candidate will also have great opportunities to work on interdisciplinary research problems with other scientists of the center.

We also expect successful applicants to do cutting-edge interdisciplinary research and work in synergy towards improving the collaborations and connections among the different areas. Ability to work and connect to more than one topic and having interdisciplinary perspective is an advantage, but disciplinary applications will be considered as well having a focus and in-depth expertise in a particular topic mentioned above.

Qualifications

- PhD in Computer Engineering, Computer Science, Electrical Engineering, or equivalent. Expertise in Quantum Physics is a plus.
- Proficiency in software programming, including Python, machine learning libraries (PyTorch/TensorFlow), code version control (git), databases (MySQL), high-performance computing (SLURM), quantum machine learning (Qirq / TensorFlow Quantum).
- Familiarity with feature engineering, representation learning, and topological data analysis.

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26